Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Audio conclusion	
Open handed problem -2	Date:25-07-2026	Enrolment No: 92400133079

Code 1:-

```python
import librosa
import soundfile as sf
import numpy as np
import  matplotlib.pyplot  as  plt
from scipy.signal import convolve

audio, sr = librosa.load(r"D:\sem-5\dsip\Haan_Karde.mp3", sr=None, mono=True)

kernel = np.array([1, 0, 1, 0, 1], dtype=np.float32)

convoluted = convolve(audio, kernel, mode="same")

convoluted /= np.max(np.abs(convoluted))

sf.write("output_convoluted.wav", convoluted, sr)

print("Saved as output_convoluted.wav")

audio /= np.max(np.abs(audio))

plt.figure(figsize=(12, 6))

plt.subplot(2, 1, 1)
plt.plot(audio)
plt.title("Original Audio")
plt.xlabel("Samples")
plt.ylabel("Amplitude")

plt.subplot(2, 1, 2)
plt.plot(convoluted)
plt.title("Convoluted Audio")
plt.xlabel("Samples")
plt.ylabel("Amplitude")

plt.tight_layout()
plt.show()
```

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